



IS222 – Database Management Systems

Semester I, 2025

Mode: Face to Face & Online (Only Online Submission Required)

Assignment 2 - Category: Group Assignment [Max of 5 Members]

Due Date: Sunday 1st June 2025

Weight: 10%

Assignments Objectives

- This assignment tests the following course learning outcomes and the associated CBOK.

CLO 3: Utilize SQL for data definition, data manipulation, and data control.

Assignment Objectives

The purpose of this assignment is to help you learn Structured Query Language and database design.

This assignment tests your SQL skills.

You will work with MySQL and write a variety of queries concerning the Data Definition Language (DDL) and the Data Manipulation Language (DML).

***Note:** The attached marking rubric on pages 7-9 will be used to assess this assignment.

Prerequisites

Each student is expected to complete this assignment in groups of four only.

Before you commence this assignment, you should have at least completed:

- Completed all required Moodle (Course) activities from Weeks 8 to 11.
- You should have covered all materials and lab exercises at least from weeks 8 to 11.

***Note:** The assignment will primarily look at the content covered from Weeks 8 to 11.

Assignment Overview

A Case Study: The University of the South Pacific – Course & Placement Database Design

Background:

The University of the South Pacific (USP) offers a range of professional short courses through its Continuing & Community Education (CCE) division. These certified programs are aimed at upskilling students and preparing them for industry opportunities. USP partners with local and regional companies that provide job openings relevant to these qualifications.

To streamline operations and improve data-driven decision-making, USP seeks to implement a relational database system to manage candidate enrollments, academic progress, and job placements.

Operational Requirements (Derived from ERD):

1. Courses and Prerequisites:

- USP offers various courses, each with a description, fee, and a qualification it leads to.
- Some courses have prerequisite courses that students must complete before enrolling.

2. Sessions and Enrollments:

- Each course runs multiple sessions, each defined by a start date and session fee.
- Students (candidates) enroll in sessions. Their enrollment date and fee payment status must be tracked.

3. Qualifications and Education Records:

- Upon completion of a course, students earn a qualification.
- These achievements are recorded in the education history, showing the qualification, candidate, and date earned.

4. Company Partnerships and Job Openings:

- USP maintains collaborations with various companies that list job openings requiring specific qualifications.
- Each job opening includes a description, availability status, and required qualification.

5. Placements and Job History:

- Students can be placed into job openings. Their placement date and total hours worked are recorded.
- USP also keeps a job history to track past placements for each candidate.

Your Task: 1

As a database designer, your role is to design a relational Entity-Relationship Diagram (ERD) based on the requirements outlined above. Your ERD should:

- Identify all relevant **entities**, **attributes**, and **relationships**.
- Clearly denote **primary keys (PK)** and **foreign keys (FK)**.
- Use appropriate **cardinality** and **participation constraints**.
- Capture the full cycle from **enrollment** to **qualification** to **job placement**.

Helpful Hint: Mapping Scenario Elements to ERD Components

As you design your Entity-Relationship Diagram (ERD) based on the scenario provided, consider how the following elements in the system might translate into entities and relationships in your design. This table offers a high-level mapping to guide your analysis:

Scenario Element

- Courses with prerequisites
- Course sessions
- Student (candidate) enrollments
- Tracking fee payment and session details
- Qualifications earned
- Partnerships with companies and job openings
- Candidate placements and history tracking

Your Task: 2

- R1. Notepad will be used to construct the three SQL batch/script files outlined below. You will run these files through the MySQL shell to see how they work. It's possible to use comments/remarks to explain what queries are written using [--].
- R2. Create a Notepad text file called GroupX_A2_CreateStructures.sql. The Xs represent your Group Name. In this file, you will write all your SQL queries to only generate the table structures as given in the solution ERD [Figure 1]. This task is part of DDL. Note the following:

Table 1: Details the requirement for the Notepad text file [GroupX_A2_CreateStructures.sql]. *X represents your Group Name

Num	Requirements
1.	When you run your script, all the tables should be created with their constraints as given in the ERD.
2.	Add comments to indicate which SQL statements correspond to which tables using --.
3.	Attribute/entity names should be similar if not the same as in the ERD. You can also add any additional useful attributes to the given tables and would not necessarily need to remove any existing ones. Consider any such changes that you introduce carefully
4.	The basic structure in the given ERD must be captured in your implementation as given. Any significant changes, such as additions of extra attributes or entities, must be clearly and very briefly explained with comments in the script where the relevant SQL statement is.
5.	You need the SQL statements to create the tables here. Any other types of statements may not be needed.
6.	Test out your script in the MySQL shell. When it runs, it is expected to create all the tables with all their required constraints without any errors. Avoid syntax errors.

R3. Create another text file in Notepad called GroupX_A2_InsertData.sql. The X represents your Group Name. In this file, you will write all your SQL statements to populate the tables created in the first step with data. This script will only be executed after the first script above. Note the following:

Table 2: Details the requirement for the Notepad text file [GroupX_A2_InsertData.sql]. *

X represents your Group Name.

Num	Requirements
1.	The data to be inserted in each table is entirely up to you but should be adding the realistic data pertinent to the scenario. Add only 10 records per table and follow any constraints given in the scenario in ERD. Also, add comments/remarks to indicate which statements correspond to which tables using --.
2.	Avoid any code that ignores any particular constraint checks when inserting data.
3.	Test out your script to see if it works without errors (syntax or other)

R4. To create the queries that will facilitate the extraction of information for reports follow the instructions given below and create a text file called GroupX_A2_Queries.sql. The Xs represent your Group Name. Write in this file, that all the queries below are based strictly on the given ERD.

Table 3: Details the requirement for the Notepad text file [GroupX_A2_Queries.sql]. *X

represents your Group Name.

No.	Requirements
1.	Add comments to indicate the query you are writing, e.g. Query 1, Query 2, etc. If the query is ambiguous, you may not get any credit for that query.

2.	Make sure your queries have no syntax errors. Avoid any gratuitous elements in your queries where possible. Test them out.
3.	Each query is to generate results based on the requirements given below:

Queries:

Write the following queries to make information extraction easier for report generation:

- Q1. The executives at The University of the South Pacific (USP) want a list of all the candidates and their qualifications. The candidates' qualifications should be sorted in ascending order on the list.
- Q2. Management would like to know the companies that have employed any of their candidates. The list should be sorted by company name in ascending order.
- Q3. To help management determine if the candidates' qualifications are in line with those that the companies a looking for a list is required of the Candidate's details, their qualifications the company's details, and the qualifications they are looking for.
- Q4. For each of the identified candidates that require further education, management will need to inform each student of the courses, when each course will start, the respective cost of each course, and the total cost of the required courses. The list is to be ordered by the candidate's last name in ascending order.
- Q5. To help finalize financial records for the 1st quarter of the year, management needs to know candidates who have not paid their fees to contact them.
- Q6. For each opening, management will need to determine the most suitable candidate/s. Each potential candidate will then be further evaluated to ensure maximum suitability. List all the companies that have current openings and the details, qualifications, and number of successful placements of candidates that could be shortlisted for these openings.

Q7. To help determine the average pay rate based on qualifications for the respective companies, management needs a list of all the companies, the qualifications they are requesting for the openings, and the average pay for openings by each company.

Submission Guidelines:

- Submit on Moodle, and upload the solutions in the A2 drop-box provided.

For Example:

Notepad: The Xs represent your Group Name.

- Create a Notepad text file called GroupX_A2_CreateStructures.sql. Notepad 2:
- Create a second text file in Notepad called GroupX_A2_InsertData.sql.
- Create a third and final file called GroupX_A2_Queries.sql.

LATE SUBMISSION = ZERO (0)

Plagiarism

- Incorrect submissions will result in loss of marks or simply a grade of zero being awarded.
- Late Submission = Grade of Zero (0) for Assignment 2.

Zero for any penalty cases thus no two assignment submissions should produce the same results. Any such case will be dealt with severely, with actions including awarding a mark of zero, forwarding the case to the Head of School, and reporting the matter to the Student Disciplinary Committee.

Assignment Queries

- Maintain contact through Moodle discussion forum named Assignment 2 Discussion Forum.
- Feel free to make an appointment via email with the Course Coordinator for any urgent matters regarding Assignment 2.

Assignment 2 Assessment Rubric

To be released soon separately.

General Comments

The End.